

Arboretum Eco-Quest

Read the living collection and solve the checkpoints before time runs out.

Win condition: Most accurate checkpoint solutions with safe field conduct.

THE SCIENCE YOU ARE USING

Observation as evidence. Leaf shape, bark texture, branching pattern, and seeds are observable traits that help narrow possibilities. Use the adaptive evidence key to record those traits, but do not treat them as proof of species. Record a species name only when a staff-verified numbered field tag, visible label, or route key confirms it.

Ecosystems in place. The arboretum is a living classroom of ecosystems. Each checkpoint asks you to observe carefully, classify what you see, and record the evidence that justifies your answer.

HOW TO BUILD AND RUN IT

- 01 Get the map and rules.** Each team gets a route map and the field-conduct rules. Stay on paths, stay together.
- 02 Solve each checkpoint.** Use the adaptive evidence key to record leaf, bark, and seed traits at each station. Name a species only from a staff-verified numbered field tag, visible label, or route key.
- 03 Collect evidence tokens.** Record the specific clue that justifies each answer and collect the token.
- 04 Keep an efficient pace.** Order your route to avoid backtracking; pace counts but accuracy counts more.
- 05 Check your answers.** Before returning, confirm each answer cites real evidence, not a guess.

Safety: Stay on route. No off-trail shortcuts. Buddy system and hydration check. Allergy: Pollen and insect-sting awareness.

MATERIALS

Clue cards	4 sets (one per team, plus 1 spare set)
Route map	4 (plus 2 spares)
Adaptive evidence key (laminated); staff-verified tags, labels, or route key confirm species names	1 per team (plus 1 spare)
Clipboards	4 (plus 1 spare)
Pencils	shared
Evidence token bags	4 (plus 1 spare), with evidence tokens for each checkpoint
QR codes	optional (open the Arboretum Explorer record)

YOUR PLAN AND DATA

Clear, well-justified answers are worth more than a fast finish: pace counts, but accuracy counts more.

Prediction (what will work best, and why): _____

Our planned route order (checkpoint numbers, to avoid backtracking): _____

Checkpoint answers (one name or ID per checkpoint): _____

The clue that justifies each answer (leaf / bark / branching / seed): _____

DEFEND YOUR ANSWERS

Which clue was most reliable for telling these trees apart, and why? _____

If you ran the route again, what is the ONE thing you would change to solve faster or more accurately? _____

HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Checkpoint accuracy	50
Evidence quality	20
Pace	20
Teamwork and conduct	10
Total	100